Hari Nivedhan P

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github.com/hari7niv | github.com/hari7niv (Portfolio) | linkedin.com/in/hari-nivedhan

Professional Summary

Software Developer building RAG-style retrieval pipelines (hybrid lexical + dense retrieval, cross-encoder reranking) and LLM inference/quantization tooling in Python, PyTorch, and Hugging Face Transformers. Backend experience across Java/Spring Boot, with REST API design, distributed systems (Kafka, Redis), and Docker. Oracle Certified Java SE 21 Developer with 300+ LeetCode problems solved in DSA.

Education

Saveetha Engineering College, Thandalam, Chennai *2024 – 2028*
B.Tech in Information Technology CGPA: 8.5 / 10
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

Skills

AI / ML: Python, PyTorch, Hugging Face Transformers, bitsandbytes (LLM.int8, NF4 quantization), LangChain, Scikit-learn, Local LLM Inference, Hybrid Retrieval (BM25 + Dense Embeddings), Reciprocal Rank Fusion, Cross-Encoder Reranking
Languages & Frameworks: Java (JDK 21), Python, JavaScript (ES6+), Spring Boot, Spring Security (JWT), MERN Stack (MongoDB, Express.js, React, Node.js), REST API Design
DevOps & Tools: Git, GitHub, Docker, Maven, Postman, Linux
Databases: MongoDB, PostgreSQL, MySQL, Redis

Professional Experience

Skill First Labs, MERN Stack Intern *July 2025*
- Engineered full-stack features across the MERN stack (MongoDB, Express.js, React, Node.js) within a production-style codebase.
- Built and integrated REST API endpoints with React front-end components, enabling end-to-end feature delivery.
- Designed MongoDB schemas and client-server architecture components, strengthening database design and API development practices.

Academic Projects

Deep Semantic Talent Retrieval Engine (RAG Pipeline) GitHub
- Built a hybrid RAG-style retrieval pipeline combining lexical (BM25S) and dense vector (BAAI/bge-small) search, fused via Reciprocal Rank Fusion to boost Stage 1 recall over either method alone.
- Added a Cross-Encoder (MiniLM) re-ranking stage to score full query-document interactions, reducing vocabulary mismatch and improving top-k precision – the same retrieve-then-rerank pattern used in production RAG systems.
- Designed a sigmoid-normalized score-fusion layer blending semantic relevance with behavioral signals (recruiter response rate, experience fit) into a single ranking score.

Model Compression & Inference Optimization Platform GitHub
- Built a config-driven LLM quantization benchmarking platform in Python, PyTorch, and Hugging Face Transformers, running FP16/INT8/INT4 local inference on Qwen3-1.7B via `bitsandbytes` with no code changes required to add new quantization techniques.
- Built a benchmarking harness measuring p50/p95/p99 latency, TTFT, throughput, and peak memory, surfacing a counterintuitive, measurement-verified finding: INT8 quantization ran 3.7x slower than FP16 due to outlier-decomposition kernel overhead, despite using more bits than INT4.
- Designed a sensitivity-guided mixed-precision search: profiled all 28 transformer blocks via fake-quantization perplexity-delta measurement, then greedily assigned per-block bit-width under a memory budget and applied it as real per-block `bitsandbytes` layer swaps, not a simulation; evaluated via Pareto-frontier (non-dominated) analysis across latency, memory, and perplexity.

High-Throughput Order Processing System GitHub
- Developed a Java/Spring Boot order processing system with multi-item orders, inventory reservation, and asynchronous fulfillment (custom thread pools, Apache Kafka, retry policies, dead-letter queues, idempotent consumers); benchmarked 3 concurrency strategies (optimistic retry, pessimistic locking, Redis distributed locking) under 50+ concurrent requests, achieving 10/10 correct reservations with zero overselling.

Certification & Achivements

- Oracle Certified Professional: Java SE 21 Developer.
- NPTEL Certification: Programming, Data Structures and Algorithms Using Python.
- Solved 350+ Data Structures & Algorithms problems on LeetCode.